

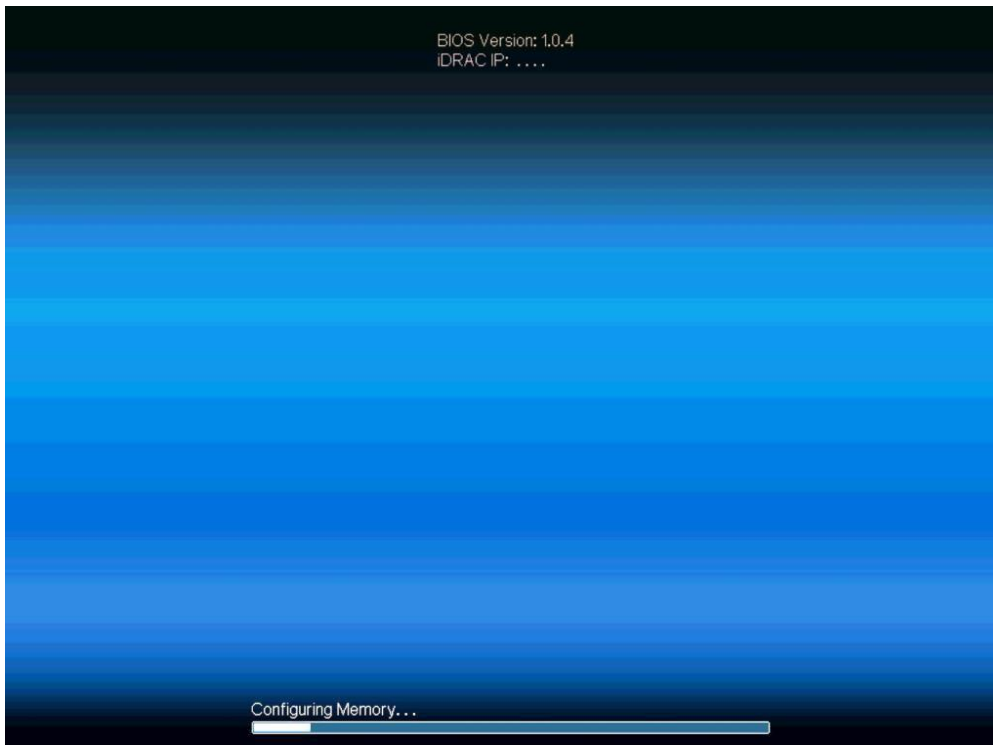


Dell™ PowerEdge™ R630 RAID configuration

Use the following procedure to configure the Avaya Dell R630 RAID controller (H730 Mini) for Avaya RAID Array configurations.

A monitor, USB keyboard and mouse are required to configure the RAID Controller.

1. Connect a monitor, USB keyboard, and mouse to HP ProLiant DL360 G9 Server.
2. Power up the server.



3. A hardware boot screen will first appear.



```
F2 = System Setup
F10 = Lifecycle Controller
F11 = Boot Manager
F12 = PXE Boot

Initializing Serial ATA devices...
Port J: HL-DT-ST DVD+/-RW GU90N

Broadcom NetXtreme Ethernet Boot Agent
Copyright (C) 2000-2014 Broadcom Corporation
All rights reserved.
Press Ctrl-S to enter Configuration Menu

PowerEdge Expandable RAID Controller BIOS
Copyright(c) 2014 LSI Corporation
Press <Ctrl><R> to Run Configuration Utility
F/W Initializing Devices 0%
```

4. Select **<F2> System Setup** when second hardware boot screen appears. Blue highlight indicates **System Setup** Menu is selected.

```
F12 = PXE Boot

Initializing Serial ATA devices...
Port J: HL-DT-ST DVD+/-RW GU90N

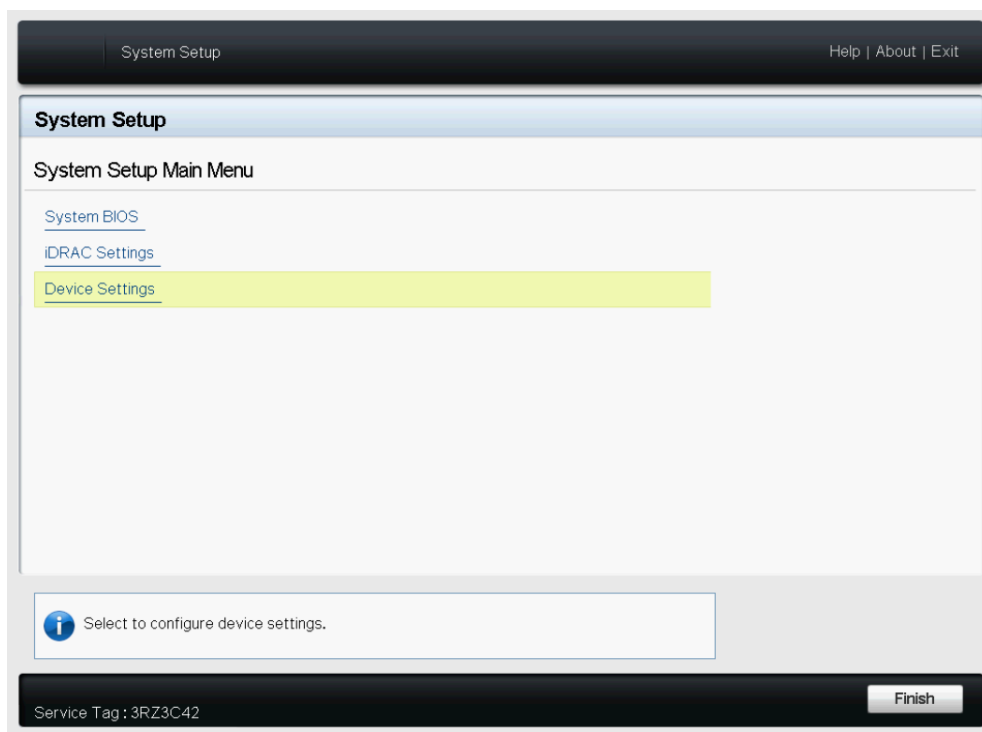
Broadcom NetXtreme Ethernet Boot Agent
Copyright (C) 2000-2014 Broadcom Corporation
All rights reserved.
Press Ctrl-S to enter Configuration Menu

PowerEdge Expandable RAID Controller BIOS
Copyright(c) 2014 LSI Corporation
Press <Ctrl><R> to Run Configuration Utility
HA -0 (Bus 3 Dev 0) PERC H730 Mini
FW package: 25.2.2-0004
```

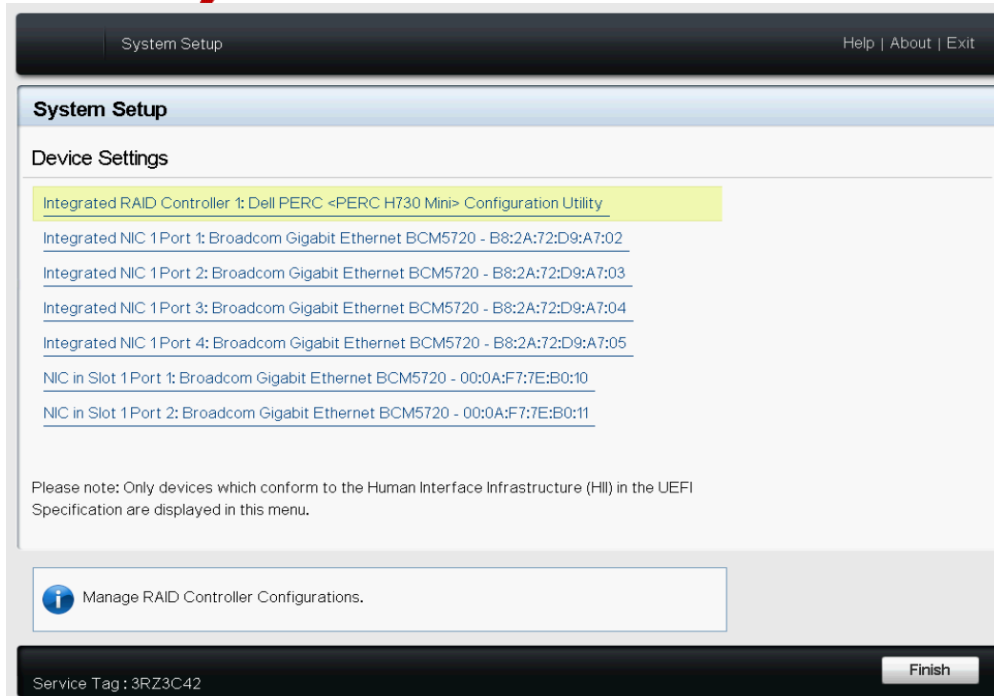
5. Server will continue hardware boot.



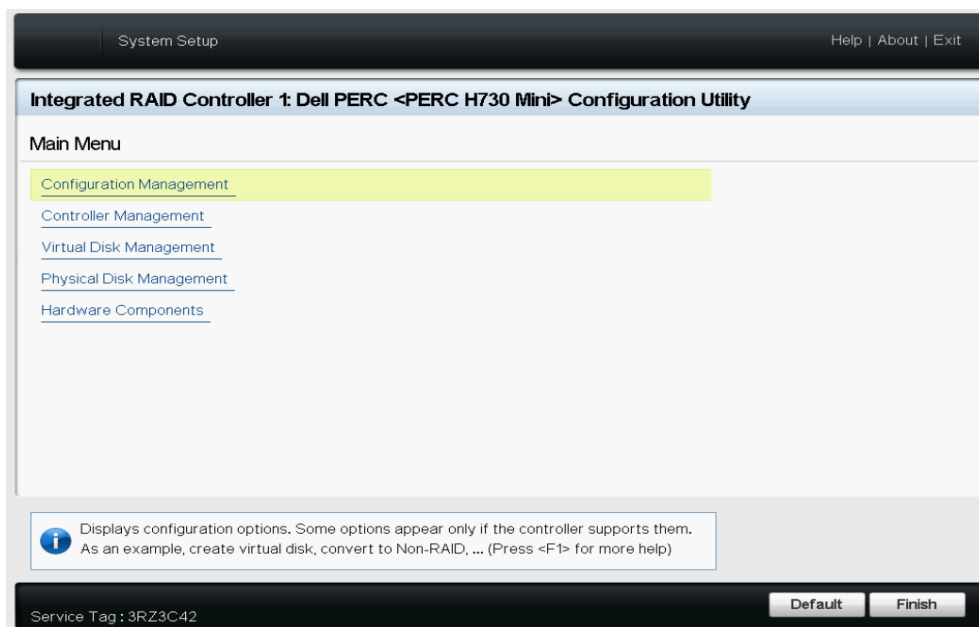
6. Final Hardware Boot proceeding to **System Setup**.



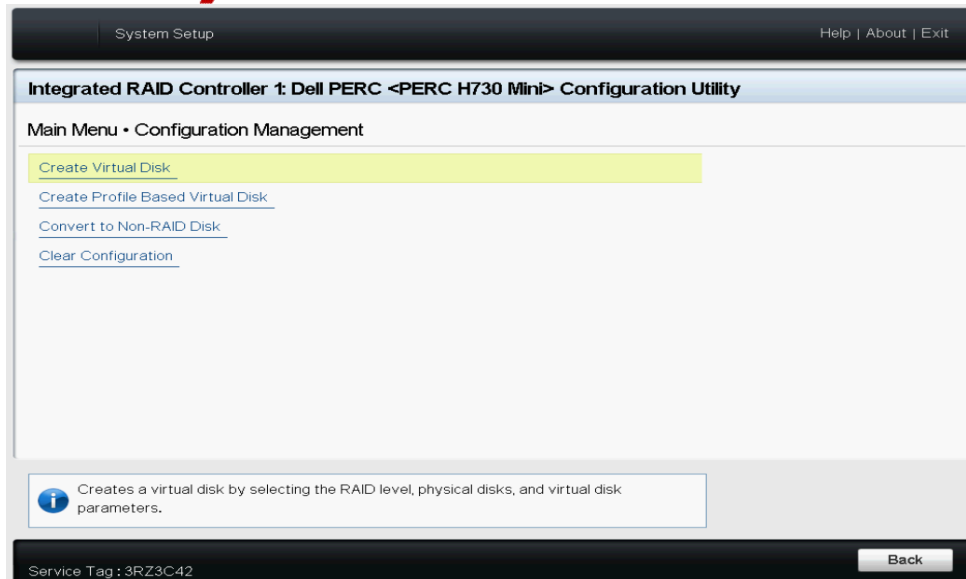
7. Select **Device Settings**.



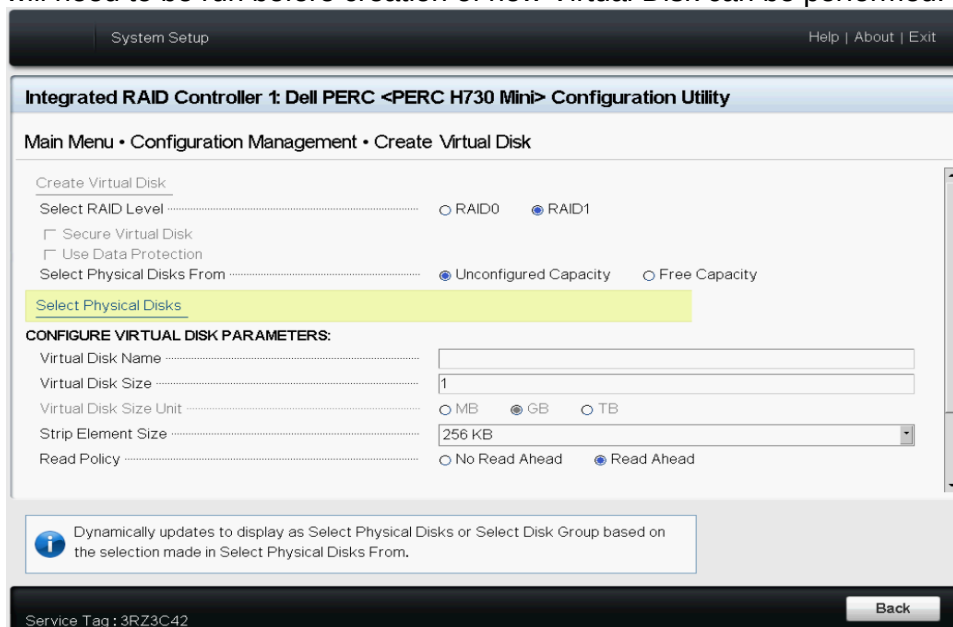
8. Select **Integrated RAID Controller 1: Dell PERC <PERC H730 Mini> Configuration Utility**.



9. Select **Configuration Management**.



10. Select **Create Virtual Disk**. If there is not an option to create a Virtual Disk then all disks in system must already be assigned to an Array. **Clear Configuration** will need to be run before creation of new Virtual Disk can be performed.



11. Select **RAID Level** required by Avaya Application. Two HDDs set for **RAID1**; Three HDDs set for **RAID5**; Four HDDs may be set for **RAID5** or **RAID10** depending on Avaya application requirements. Leave the **Unconfigured Capacity** button selected and **Virtual Disk Name** blank. Change **Strip Element Size** to 256 KB. Scroll down and select **Fast** button for **Default Initialization**. Now press the **Select Physical Disks** link shown on screen below.



System Setup Help | About | Exit

Integrated RAID Controller 1: Dell PERC <PERC H730 Mini> Configuration Utility

Main Menu • Configuration Management • Create Virtual Disk

Select Physical Disks From ☒ Unconfigured Capacity ☐ Free Capacity

[Select Physical Disks](#)

CONFIGURE VIRTUAL DISK PARAMETERS:

Virtual Disk Name

Virtual Disk Size

Virtual Disk Size Unit ☐ MB ☒ GB ☐ TB

Strip Element Size

Read Policy ☐ No Read Ahead ☒ Read Ahead

Write Policy ☒ Write Back ☐ Write Through ☐ Force Write Back

Disk Cache ☒ Default ☐ Enable ☐ Disable

Default Initialization ☐ No ☒ Fast ☐ Full

[Create Virtual Disk](#)

Dynamically updates to display as Select Physical Disks or Select Disk Group based on the selection made in Select Physical Disks From.

Service Tag : 3RZ3C42 [Back](#)

System Setup Help | About | Exit

Integrated RAID Controller 1: Dell PERC <PERC H730 Mini> Configuration Utility

Main Menu • Configuration Management • Create Virtual Disk • Select Physical Disks

[Apply Changes](#)

Select Media Type ☐ SSD ☒ HDD ☐ Both

Select Interface Type ☐ SAS ☐ SATA ☒ Both

Logical Sector Size ☐ 512 B ☐ 4 KB ☒ Both

CHOOSE UNCONFIGURED PHYSICAL DISKS:

☒ Physical Disk 00:01:00: HDD, SAS, 278GB, Ready, (512B)

☒ Physical Disk 00:01:01: HDD, SAS, 278GB, Ready, (512B)

[Check All](#)

[Uncheck All](#)

[Apply Changes](#)

Submits the changes made to the entire form.

Service Tag : 3RZ3C42 [Back](#)

12. Leave the top three lines as default and select the drives that are required for the Array being created. **Check All** physical disks unless configuring more than one Virtual Disk. Press **Apply Changes** once physical disks have been checked.



System Setup About | Exit

Integrated RAID Controller 1: Dell PERC <PERC H730 Mini> Configuration Utility

Main Menu • ... • Create Virtual Disk • Select Physical Disks • Success

The operation has been performed successfully.

[OK](#)

Service Tag : 3RZ3C42

13. Select **OK** to confirm operation was successful. One more step must be performed to create the Virtual Disk!

System Setup Help | About | Exit

Integrated RAID Controller 1: Dell PERC <PERC H730 Mini> Configuration Utility

Main Menu • Configuration Management • Create Virtual Disk

[Create Virtual Disk](#)

Select RAID Level ☐ RAID0 ☒ RAID1

☐ Secure Virtual Disk

☐ Use Data Protection

Select Physical Disks From ☒ Unconfigured Capacity ☐ Free Capacity

[Select Physical Disks](#)

CONFIGURE VIRTUAL DISK PARAMETERS:

Virtual Disk Name

Virtual Disk Size

Virtual Disk Size Unit ☐ MB ☒ GB ☐ TB

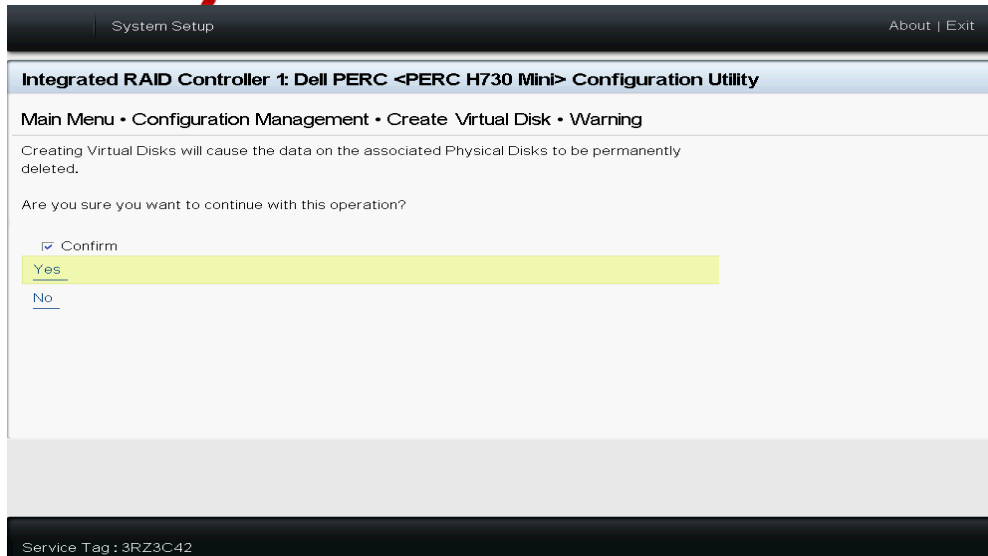
Strip Element Size

Read Policy ☐ No Read Ahead ☒ Read Ahead

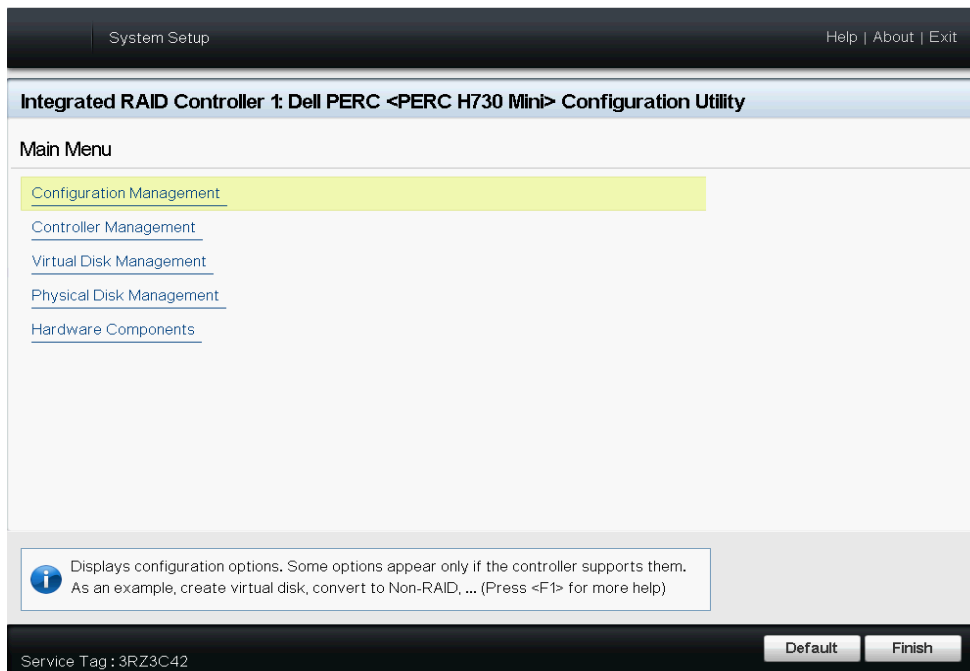
Submits the changes made to the entire form and creates a virtual disk with the specified parameters.

Service Tag : 3RZ3C42 [Back](#)

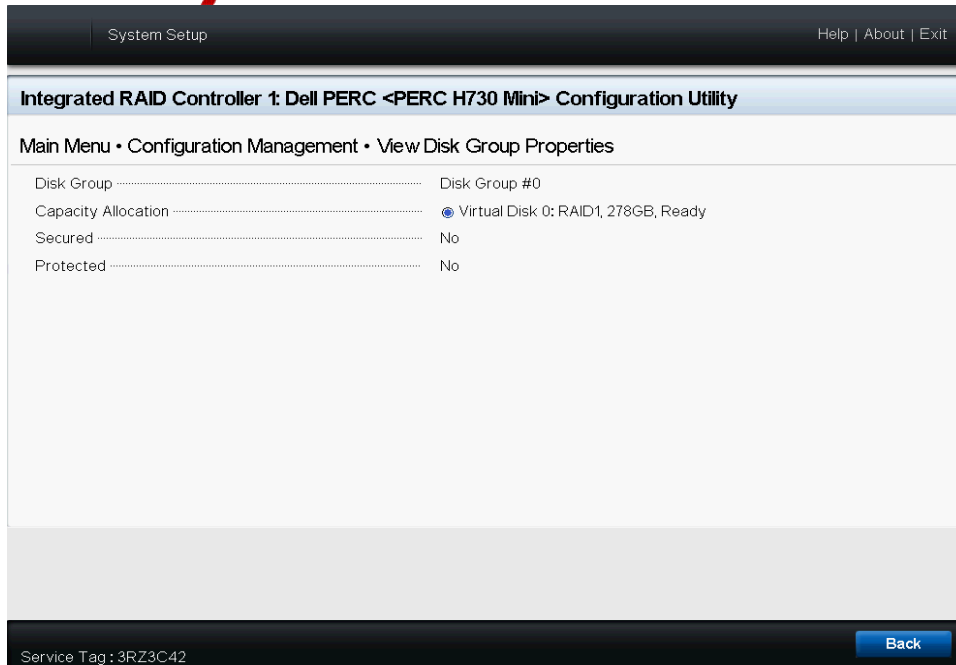
14. Now select **Create Virtual Disk**. (This link goes active once physical disks have been assigned.)



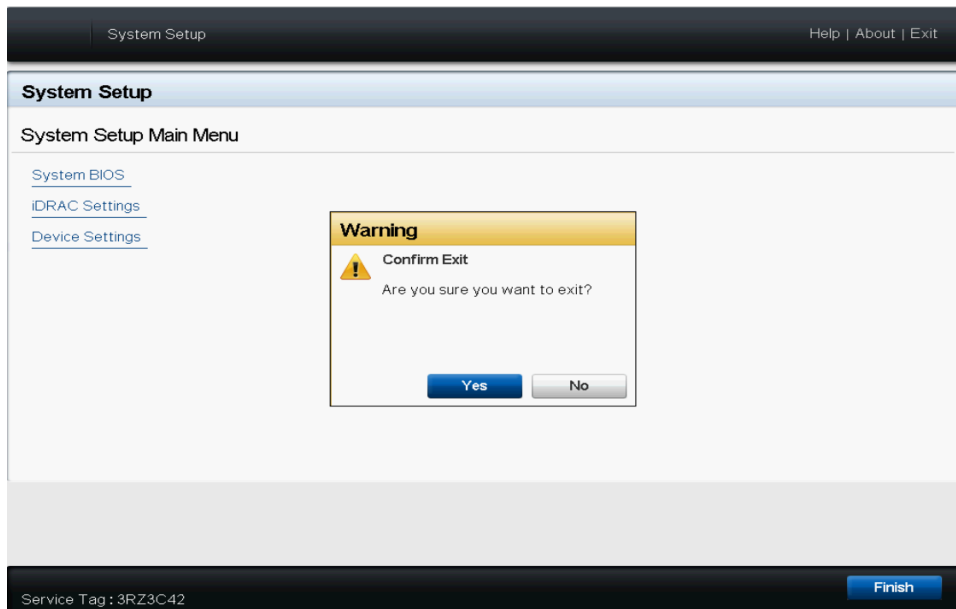
15. A confirmation screen appears. Check **Confirm** and click **Yes**. Click OK on next screen. Virtual disk should have been created. Select **Back** and **Back** again to view new Virtual Disk.



16. Now select **Configuration Management/View Disk Group Properties**.



17. The display should show the Virtual Drive just created. Select **Back**.
18. Select **Back** and then **Finish**.
19. Select **Finish** and then **Finish** again.



20. **Confirm Exit** by selecting **Yes**. Virtual Drive is now available for software installation.



Supplementary RAID Array configuration size example screens:

System Setup Help | About | Exit

Integrated RAID Controller 1: Dell PERC <PERC H730 Mini> Configuration Utility

Main Menu • Configuration Management • Create Virtual Disk

[Create Virtual Disk](#)

Select RAID Level: RAID10

☐ Secure Virtual Disk
☐ Use Data Protection

Select Physical Disks From: ☒ Unconfigured Capacity ☐ Free Capacity

[Select Physical Disks](#)

CONFIGURE VIRTUAL DISK PARAMETERS:

Virtual Disk Name:

Virtual Disk Size: 557

Virtual Disk Size Unit: ☐ MB ☒ GB ☐ TB

Strip Element Size: 256 KB

Read Policy: ☐ No Read Ahead ☒ Read Ahead

Dynamically updates to display as Select Physical Disks or Select Disk Group based on the selection made in Select Physical Disks From.

Service Tag : 3RZ3C42 [Back](#)

RAID10 – 4x300GB => 557GB Virtual Drive Size

System Setup Help | About | Exit

Integrated RAID Controller 1: Dell PERC <PERC H730 Mini> Configuration Utility

Main Menu • Configuration Management • Create Virtual Disk

[Create Virtual Disk](#)

Select RAID Level: RAID5

☐ Secure Virtual Disk
☐ Use Data Protection

Select Physical Disks From: ☒ Unconfigured Capacity ☐ Free Capacity

[Select Physical Disks](#)

CONFIGURE VIRTUAL DISK PARAMETERS:

Virtual Disk Name:

Virtual Disk Size: 836

Virtual Disk Size Unit: ☐ MB ☒ GB ☐ TB

Strip Element Size: 256 KB

Read Policy: ☐ No Read Ahead ☒ Read Ahead

Dynamically updates to display as Select Physical Disks or Select Disk Group based on the selection made in Select Physical Disks From.

Service Tag : 3RZ3C42 [Back](#)

RAID5 – 4x300GB => 836GB Virtual Drive Size



System Setup Help | About | Exit

Integrated RAID Controller 1: Dell PERC <PERC H730 Mini> Configuration Utility

Main Menu • Configuration Management • Create Virtual Disk

Create Virtual Disk

Select RAID Level: ☐ RAID0 ☐ RAID1 ☒ RAID5

☐ Secure Virtual Disk
☐ Use Data Protection

Select Physical Disks From: ☒ Unconfigured Capacity ☐ Free Capacity

[Select Physical Disks](#)

CONFIGURE VIRTUAL DISK PARAMETERS:

Virtual Disk Name:

Virtual Disk Size:

Virtual Disk Size Unit: ☐ MB ☒ GB ☐ TB

Strip Element Size:

Read Policy: ☐ No Read Ahead ☒ Read Ahead

Submits the changes made to the entire form and creates a virtual disk with the specified parameters.

Service Tag : 3RZ3C42 Back

RAID5 – 3x300GB => 557GB Virtual Drive Size